

Programming Exercises 1

1. Draw a flowchart or write pseudocode to represent the logic of a program that allows the user to enter a value. The program divides the value by 2 and outputs the result.

Vẽ sơ đồ hoặc viết mã giả để thể hiện logic của chương trình cho phép người dùng nhập một giá trị. Chương trình chia giá trị đó cho 2 và in ra kết quả.
2. Draw a flowchart or write pseudocode to represent the logic of a program that allows the user to enter a value for one edge of a cube. The program calculates the surface area of one side of the cube, the surface area of the cube, and its volume. The program outputs all the results.
3. Draw a flowchart or write pseudocode to represent the logic of a program that allows the user to enter values for a salesperson's base salary, total sales, and commission rate. The program computes and outputs the salesperson's pay by adding the base salary to the product of the total sales and commission rate.
4. A consignment shop accepts a product for sale and sets an initial price. Each month that the item doesn't sell, the price is reduced by 20 percent. When the item sells, the item's owner receives 60 percent of the sale price, and the shop gets 40 percent. Draw a flowchart or write pseudocode to represent the logic of a program that allows the user to enter an original product price. The output is the sale price, the owner's cut, and the shop's cut each month for the first three months the item is on sale.
5. A mobile phone app allows a user to press a button that starts a timer that counts seconds. When the user presses the button again, the timer stops. Draw a flowchart or write pseudocode that accepts the elapsed time in seconds and displays the value in minutes and seconds. For example, if the elapsed time was 130 seconds, the output would be 2 minutes and 10 seconds.